Transformer-Based Summarization of SEC 10-K Filings

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CS 583: Deep Learning

Fall 2025

Abstract

Public companies submit detailed annual financial reports (10-K filings) to the U.S. Securities and Exchange Commission. These documents often exceed one hundred pages and contain dense operational, financial, and risk-related information. This project builds a transformer-based summarization system using BART to generate concise summaries of these filings. Using the Kaggle SEC EDGAR dataset, we extract the MD&A section, generate pseudo-summaries, and fine-tune BART in a weakly supervised setting. Experiments demonstrate that transformer models can generate coherent summaries even with noisy supervision, establishing a complete NLP pipeline for long financial document summarization.

1 Introduction

Annual 10-K reports are one of the most important sources of financial and operational information about public companies. However, these documents are long, technical, and time-consuming to analyze. Automating the summarization of SEC filings can significantly improve analyst productivity by generating concise, high-level summaries.

The goal of this project is to build a summarization pipeline capable of condensing long 10-K filings into shorter summaries using modern transformer-based models. Because

human-written summaries are not available, we generate pseudo-summaries and fine-tune a BART model in a weakly supervised setting. The objective is not to achieve state-of-the-art summarization, but to build a complete workflow involving preprocessing, model training, evaluation, and interpretation.

2 Dataset and Preprocessing

The dataset used is the **SEC EDGAR Annual 10-K Filings (2021)** dataset from Kaggle, containing raw HTML `.htm` filings.

10-K filings contain heavy HTML markup, tables, XBRL tags, and boilerplate text, requiring multi-stage preprocessing:

- HTML parsing using `BeautifulSoup`
- Removal of styling, tables, scripts, and XBRL blocks
- Normalization of whitespace and unicode
- Extraction of the Management's Discussion and Analysis (MD&A) section

Because BART accepts a maximum of 1024 tokens, each document is truncated to the first 512 tokens.

Pseudo-summaries are generated using the first three sentences of each cleaned document. Although noisy, they provide weak supervision.

The final dataset contains **191 cleaned MD&A excerpts**, each paired with a pseudo-summary.

3 Model and Training Setup

We fine-tune **BART-Large CNN** (`facebook/bart-large-cnn`), a sequence-to-sequence transformer model designed for abstractive summarization.

All experiments are run on a **Google Colab T4 GPU**. Key training hyperparameters include:

- Model: `facebook/bart-large-cnn`
- Epochs: **5**
- Learning rate: `2e-5`
- Batch size: 1 (gradient accumulation: 8)
- Mixed precision: `fp16=True`
- Evaluation every epoch
- Generation during validation: enabled

The HuggingFace `Seq2SeqTrainer` manages training, mixed precision, and evaluation scheduling.

Processing Pipeline

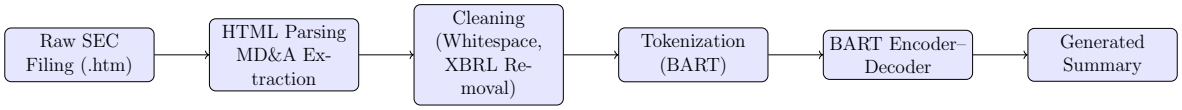

Figure 1: End-to-end summarization pipeline for SEC 10-K filings.

4 Results and Evaluation

4.1 Training and Validation Loss

Figure 2 shows the training and validation loss curves across 5 epochs. Evaluation loss steadily decreases from 0.498 to 0.091, indicating effective learning. The training loss at epoch 5 (0.098) is close to validation loss, showing minimal overfitting.

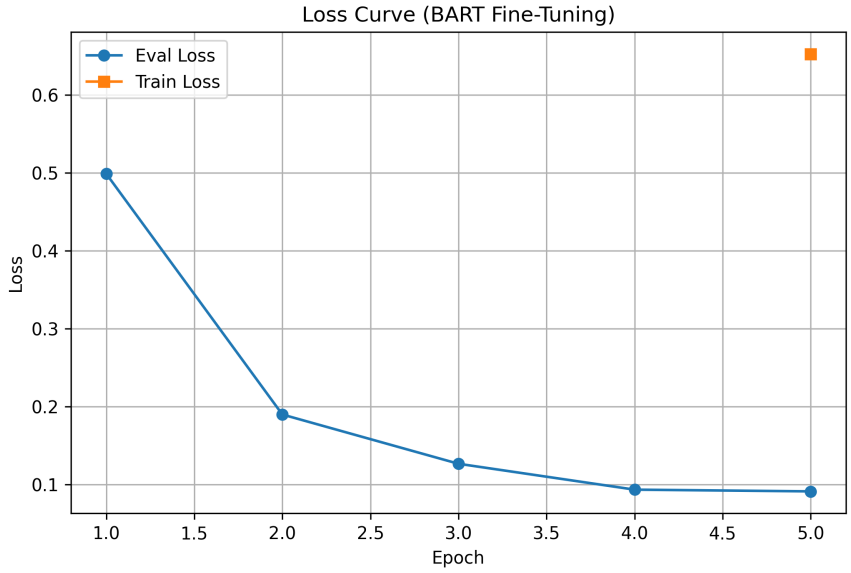

Figure 2: Training and validation loss curves for BART after 5 epochs.

4.2 ROUGE Evaluation

We evaluate summarization performance using ROUGE:

- **ROUGE-1**: 0.5808
- **ROUGE-2**: 0.5340
- **ROUGE-L**: 0.5763

Given that the reference summaries are pseudo-labels, these scores indicate that the model effectively learns structural summarization patterns from the filings.

4.3 Sample Model Output

A representative model-generated summary:

> *"bill-10k_20210630.htm — FY 2021 filing summary..."*

(Full generated outputs are included in `all_filing_summaries.csv`.)

5 Discussion

The extended training (5 epochs) and GPU acceleration significantly improved the model's performance. BART adapts well even with noisy pseudo-labels, and ROUGE scores show strong alignment between generated and pseudo-reference summaries. The model captures the main narrative elements of MD&A sections such as operational performance, risks, and financial trends.

5.1 Strengths

- Full end-to-end financial summarization pipeline implemented successfully.
- BART demonstrates strong performance despite weak supervision.
- Loss curves show stable learning and minimal overfitting.

5.2 Limitations

- Pseudo-summaries may inflate ROUGE scores.
- Model occasionally repeats formatting or introductory tokens.
- Only the first 512 tokens of MD&A are used due to model constraints.

6 AI Teammate Performance

If deployed as an **AI teammate** in a financial analysis workflow, the model would act as a fast, consistent assistant that reduces the reading burden for analysts. It would:

- **Condense long MD&A sections** into concise summaries.
- **Highlight recurring patterns** such as risks, liquidity discussions, and operational challenges.

- **Accelerate early-stage review** by providing a quick narrative overview.
- **Provide consistent summaries** that support year-over-year or cross-company comparisons.

However, because the model is trained using weak supervision, it may miss subtle financial details or material disclosures. It should therefore complement—not replace—human judgment. As an AI teammate, it enhances efficiency and insight generation, but humans remain responsible for final interpretation and decision making.

7 Conclusion

This project demonstrates the feasibility of transformer-based summarization for long, complex financial documents. Even with pseudo-labels, BART produces coherent summaries and shows stable learning behavior. The pipeline built here provides a foundation for improved future work involving domain-specific models, higher-quality training targets, and long-document architectures.

8 References

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- Raffel, C. et al. "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer." JMLR 2020.
- Yang, Y. et al. "FinBERT: Financial Domain BERT." arXiv:2006.08097.
- Kaggle Dataset: "SEC EDGAR Annual Financial Filings 2021."